FIRST PRE-DEFENSE • 24 OCTOBER 2026

Intelligent Intersection Traffic Management

Development of an adaptive, simulation-led traffic control system

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Github : https://github.com/Tompi-dev/intelligent-intersection-management
YouTrack: https://tomiris.youtrack.cloud/projects/IIM/issues



Project proposal

From fixed schedules to measurable, adaptive control

PROBLEM

Fixed-time plans cannot react to short-term demand, queues or incidents.

AIM

Develop and evaluate an adaptive intersection controller in simulation.

USERS

Traffic engineers, researchers and municipal operators.

OUTPUTS

Controller, CV pipeline prototype, experiment package and technical report.

BOUNDARY Single-intersection simulation; no live-road deployment, safety certification or citywide optimization.

Evidence of relevance

A transparent ledger: what is known, what must still be validated

VERIFIED PROJECT FACTS

- Fixed-time control is the baseline
- Simulation is the evaluation environment
- No experimental results exist yet

ASSUMPTIONS TO TEST

- Queue reduction matters most to users
- Camera feeds are sufficiently observable
- One controller can generalize across demand

INTERVIEW PLAN → traffic engineer · road user · municipal operations

INTERVIEWS PENDING

Literature review & research gap

Three mature research streams — one integration challenge



01 VEHICLE DETECTION

Real-time accuracy trades off with edge compute cost and adverse weather robustness.

02 MULTI-OBJECT TRACKING

Identity continuity enables counts, trajectories and turning-movement estimates.

03 ADAPTIVE SIGNAL CONTROL

Rule-based and learning approaches require comparable demand scenarios and safety constraints.

GAP A reproducible, end-to-end bridge from perception outputs to a defensible simulated controller benchmark.

Existing systems comparison

Selection criteria emphasize reproducibility and programmable control

SYSTEM	ACCESS	CONTROL SUPPORT	INTEGRATION / API	PROJECT FIT
SUMO	Open source	Strong	TraCI / Python	HIGH — baseline
CityFlow	Open source	Strong	Python API	HIGH — benchmark
PTV Vissim	Commercial	Strong	COM / scripting	MEDIUM — license
Aimsun Next	Commercial	Strong	API / SDK	MEDIUM — license

Initial preference: SUMO + TraCI. Final choice remains contingent on prototype integration and benchmark evidence.

Technology decisions & alternatives

Initial choices are hypotheses — not benchmark results

DETECTION	Lightweight one-stage detector	Two-stage / classical CV	INITIAL
TRACKING	Association-based tracker	Motion-only / appearance-based	INITIAL
STORAGE	SQLite + experiment files	PostgreSQL / object store	INITIAL
INTERFACE	CLI + config files	Web dashboard	IMPLEMENTED CORE

Benchmark before locking: accuracy • latency • integration effort • reproducibility

Requirements & proposed architecture

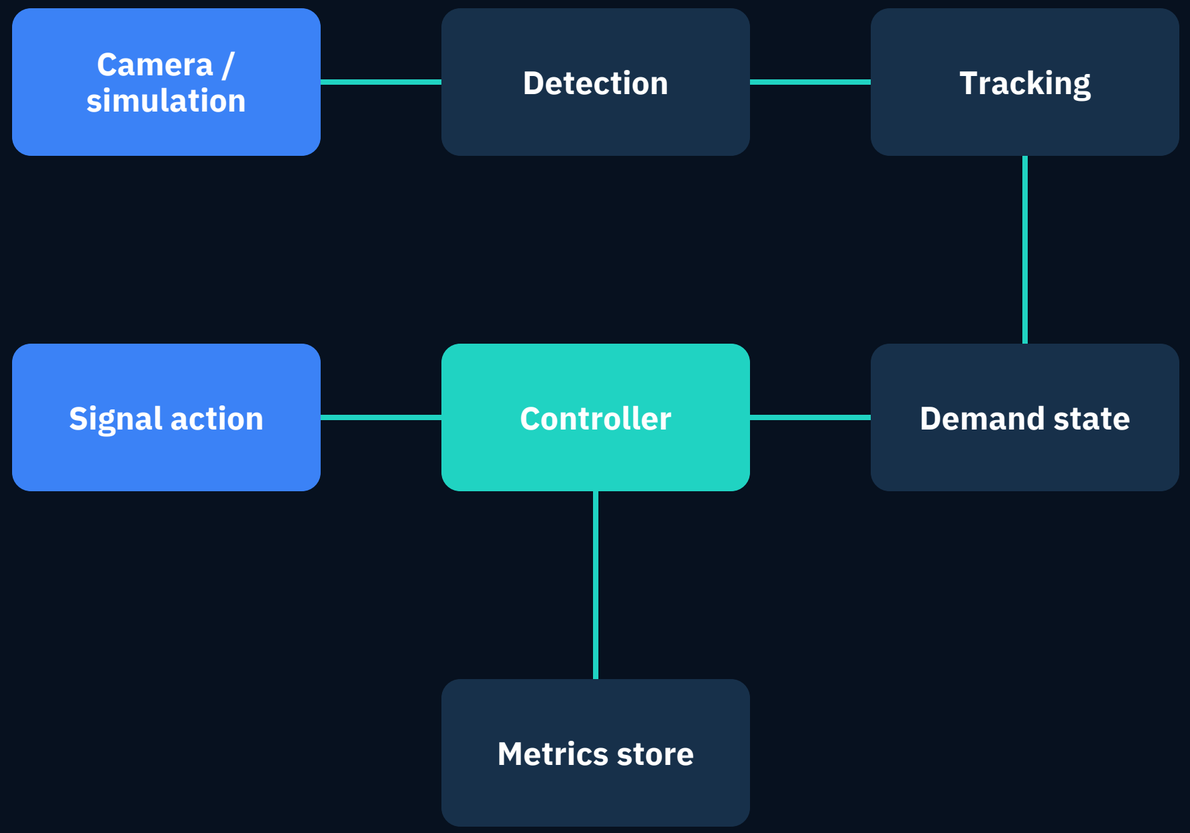
Target architecture — interfaces and acceptance criteria remain editable

FUNCTIONAL

Ingest detections
Track vehicles
Estimate demand
Select signal phase
Log each decision

NON-FUNCTIONAL

Reproducible runs
Configurable scenarios
Traceable outputs
Safe phase constraints



ACCEPT Controller completes seeded scenarios, respects phase constraints, and exports comparable metrics.

Prototype: implemented vs planned

Status labels distinguish working software from design intent

IMPLEMENTED

CLI CONTROLLER

Configuration-driven runs
Phase-selection logic
Console outputs and basic logs

PLANNED

PERCEPTION + UI

Computer-vision pipeline
Tracker integration
Operator dashboard

UI MOCKUPS ARE NON-FUNCTIONAL

Demo claim: software path only. No claim of validated traffic improvement.

Verification & evaluation plan

Planned tests only — no results exist yet

UNIT

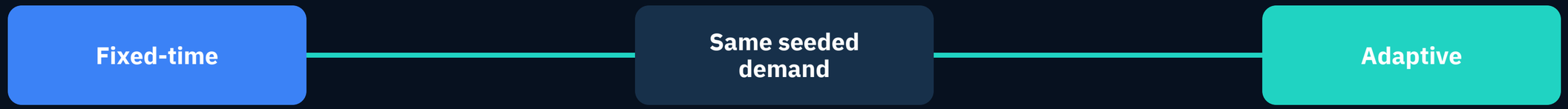
Phase logic, constraints, metric calculations

INTEGRATION

Detector → tracker → controller contracts

SIMULATION

Fixed-time vs adaptive under identical demand



PAIRED DESIGN Compare queue length, delay and throughput per seed; report uncertainty and failure cases.

Project management & team workflow

Proposed operating model — completion claims require repository evidence

REPOSITORY

GitHub
branches + pull requests

TRACKING

YouTrack
issues + acceptance criteria

REVIEW

Peer review
CI checks + experiment logs

MILESTONES

Prototype → benchmark → report

PROPOSED RESPONSIBILITIES

Perception lead · control & simulation lead · validation & documentation lead

Status source of truth: merged code, linked issue and reproducible run artifact.

Risk register & supervisor review

Prioritize uncertainty that can invalidate the evaluation

CV reliability	High	Weather / occlusion benchmark; simulation fallback
Simulator integration	High	Thin adapter; early end-to-end spike
Scope growth	Medium	Single intersection; explicit change control
Weak evidence	High	Paired seeds; baselines; uncertainty reporting

SUPERVISOR REVIEW PENDING • validate research question • approve metrics • confirm scope

Readiness summary & next steps

Honest status: a credible plan with an implemented controller core

READY

Problem framing
Architecture
CLI controller

PENDING

Stakeholder interviews
Supervisor review
Benchmarks

NEXT

Freeze scenarios
Integrate simulator
Run paired tests

PRE-DEFENSE DELIVERABLES

Proposal · literature synthesis · architecture · prototype demo · evaluation plan

Decision requested: approve scope and evaluation protocol.